

API Automation Assessment Report

Project Information

Project Name: Broker

Participant: Jamesy Joseph

Date: 29/06/2026

Branch: Assessment2_jamesy29Jun2026_broker

Assessment: API Automation using Rest Assured

1. Objective & User Story

Objective

The objective of this automation project is to validate the core REST API functionality of the application using Rest Assured. Designed to be scalable, maintainable and reusable using modern API automation best practices.

The automated test suite validates critical API workflows including authentication, resource retrieval, update and deletion. It ensures that API responses behave as expected.

User Story

As a QA Engineer, I want to validate the Booking API using invalid inputs and verify delete functionality.

2. Flow Under Test

Scenario 1: Validate the Credentials

1. Send POST request to Authentication endpoint.
2. Provide valid username and wrong password.
3. Verify response status code (Status Code – 200)
4. Response contains "Bad credentials".

Scenario 2: Retrieve Booking by ID - 999999

1. Send GET request using Booking ID.
2. Verify response status code (404).

Scenario 3: Generate Authentication Token

1. Send POST request to Authentication endpoint.
2. Provide valid username and password.
3. Verify response status code.
4. Verify authentication token is generated.

Scenario 4: Update Existing Booking

1. Generate authentication token.
2. Send PATCH request.
3. Verify response status code 200.
4. Verify updated values ("firstname": "Michael").

Scenario 5: Delete Existing Booking

1. Generate authentication token.
2. Send DELETE request.
3. Verify response status code 201.
4. Verify successful deletion response.

Scenario 6: Retrieve Booking by ID-(Deleted Booking ID)

1. Send GET request using deleted Booking ID.
2. Verify response status code (404).

3. Environment & Setup

Application Details

Item - Value

Application URL - Configured through .env

Framework - Rest Assured

Language - Java 22

Build Tool - Maven

Test Runner - JUnit 5

Design Pattern - API Client

Schema Validation - JSON Schema Validator

Authentication - Token Based

OS - Windows 11

Environment Variables

Stored in: .env file

BASE_URL = https://restful-booker.herokuapp.com

USER_NAME = admin

PASSWORD = password123

WRONG_PASSWORD = wrongPassword

BASE_PATH = /booking

AUTH_PATH = /auth

Execution Command

mvn clean test

4. Framework Design

The framework follows reusable API automation design principles.

Folder Structure

src

1) test

Java/com/ api_testing/

tests

MainTest.java

models

AuthRequest.java

AuthResponse.java

BookingRequest.java

BookingResponse.java

BookingDates.java

BookingGetResponse.java

specs

ReqSpec.java

ResSpec.java

support

Config.java

2) **.env**

3) **pom.xml**

Components

Tests

Contains all API test scenarios.

File: MainTest.java

Models

Contains Record classes used for serialization and deserialization.

Files: AuthRequest, AuthResponse, BookingRequest, BookingResponse, BookingDates.java, BookingGetResponse.java

Specifications

Contains reusable Request Specifications and Response Specifications.

File: ReqSpec.java, ResSpec.java

Support

Contains: Config.java

5. Test Scenarios Covered

Positive Scenarios

Test Case ID – Description – Status

TC_API_001 - Generate Authentication Token - Passed

TC_API_002 - Retrieve Booking ID-99999 - Passed

TC_API_003 - Update Booking - Passed

TC_API_004 - Delete Booking - Passed

Negative Scenarios

Test Case ID – Description – Status

TC_API_005 - Invalid Authentication - Passed

TC_API_006 - Retrieve Invalid Booking ID - Passed

6. Automation Best Practices Used

Rest Assured Framework, JUnit 5, Request Specification Pattern, Response, Specification Pattern, Environment Variables (.env), Record, Reusable Test Utilities, Token-Based Authentication, Centralized Configuration, Assertions for Status Codes, Clean and Maintainable Framework Structure, Reusable Payload Models, Maven Dependency Management

7. Execution Result

Metric - Value

Total Tests - 6

Passed - 6

Failed - 0

Skipped - 0

8. Results

Test 1: Generate Authentication Token

Successfully generated token

Test 2: Retrieve Booking by ID - 999999

Successfully failed booking ID - 404

Test 3: Retrieve Booking by ID

Successfully retrieved booking details and validated response content.

Test 4: Update Booking

Successfully updated booking information and verified modified fields.

Test 5: Delete Booking

Successfully deleted booking and verified successful response.

Test 6: Retrieve Booking by ID -Deleted ID

Successfully failed booking ID - 404

9. Conclusion

The Rest Assured API automation framework successfully validates the critical functionality of the application APIs.

Key achievements:

- Authentication workflow validated.
- Read, Update, Delete operations validated.
- Positive and Negative API scenarios implemented.
- Environment-based configuration implemented using .env.
- Reusable Request and Response Specifications implemented.
- Clean and maintainable framework structure followed.
- API automation best practices applied.
- Test execution completed successfully.